



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: GRID-INDIA/NLDC/SO/Daily PSP Report

दिनांक: 09.12.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई -400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु -560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 08.12.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-38(1) के प्रावधान के अनुसार, दिनांक 8-दिसंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 38(1) of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 08.12.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day

Date of Reporting: 09-Dec-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 19:00 hrs; from RLDCs)	50291	58724	40909	19623	2479	172026
Peak Shortage (MW)	0	0	0	0	0	0
Energy Met (MU)	1126	1448	1005	408	46	4033
Hydro Gen (MU)	114	31	66	12	14	238
Wind Gen (MU)	27	50	28	-	-	105
Solar Gen (MU)*	150.12	91.93	112.80	2.24	1.05	358
Energy Shortage (MU)	0.37	0.00	0.00	0.00	0.00	0.37
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	60966	71700	47985	20388	2631	200033
Time Of Maximum Demand Met	10:09	09:58	09:35	17:46	17:20	09:58

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.036	0.00	0.58	3.28	3.85	77.28	18.86

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	8300	0	145.0	64.6	-2.6	409	0.00
	Haryana	7207	0	135.2	103.4	-2.3	323	0.00
	Rajasthan	17123	0	316.6	104.3	-5.3	708	0.00
	Delhi	3811	0	67.4	67.2	-1.1	230	0.00
	UP	17454	0	317.7	96.1	-0.7	520	0.00
	Uttarakhand	2216	0	39.9	28.7	-0.2	124	0.00
	HP	1948	0	34.5	27.7	-0.6	306	0.00
	J&K(UT) & Ladakh(UT)	2873	0	61.6	57.3	-0.9	118	0.37
	Chandigarh	190	0	3.6	3.6	-0.1	34	0.00
WR	Railways_NR ISTS	203	0	4.2	3.8	0.4	36	0.00
	Chhattisgarh	4545	0	94.2	44.1	0.6	280	0.00
	Gujarat	20605	0	397.5	180.6	0.0	545	0.00
	MP	17908	0	332.6	204.8	-3.3	614	0.00
	Maharashtra	25830	0	548.3	207.4	-6.3	823	0.00
	Goa	640	0	14.4	13.5	0.4	42	0.00
	DNHDDPDCL	1248	0	28.9	28.9	0.0	31	0.00
	AMNSIL	892	0	19.8	9.5	0.3	308	0.00
	BALCO	529	0	12.6	12.7	-0.1	4	0.00
SR	RIL JAMNAGAR	0	0	0.0	0.0	0.0	0	0.00
	Andhra Pradesh	9637	0	192.6	47.7	-0.9	479	0.00
	Telangana	10879	0	215.0	126.3	1.0	835	0.00
	Karnataka	11105	0	209.1	54.5	-0.1	457	0.00
	Kerala	3973	0	78.0	62.6	1.1	283	0.00
	Tamil Nadu	14107	0	301.5	164.1	-3.5	601	0.00
	Puducherry	348	0	8.6	8.3	-0.2	26	0.00
ER	Bihar	4638	0	85.2	73.2	0.0	135	0.00
	DVC	3106	0	66.5	-32.7	0.4	394	0.00
	Jharkhand	1965	0	39.6	27.7	1.2	172	0.00
	Odisha	4489	0	88.4	21.1	-1.4	393	0.00
	West Bengal	6207	0	126.9	15.2	-2.7	285	0.00
	Sikkim	92	0	1.4	1.5	-0.1	12	0.00
	Railways_ER ISTS	13	0	0.2	0.2	0.0	3	0.00
NER	Arunachal Pradesh	149	0	3.1	3.0	0.0	11	0.00
	Assam	1427	0	24.4	18.7	0.0	125	0.00
	Manipur	258	0	3.8	3.7	0.1	44	0.00
	Meghalaya	344	0	6.5	6.2	-0.1	30	0.00
	Mizoram	126	0	2.5	1.8	0.0	22	0.00
	Nagaland	154	0	2.5	2.4	-0.1	24	0.00
	Tripura	211	0	3.1	4.5	-0.8	22	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	-4.6	-0.2	-12.0	-15.3
Day Peak (MW)	-407.0	-159.4	-550.0	-766.2

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	199.2	-154.8	141.7	-191.9	5.8	0.0
Actual(MU)	193.7	-155.9	139.1	-189.0	3.5	-8.5
O/D/U/D(MU)	-5.4	-1.1	-2.6	2.9	-2.3	-8.5

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	5620	9009	8068	3838	551	27086	50
State Sector	7963	9713	6299	2599	353	26926	50
Total	13583	18722	14367	6437	904	54012	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	633	1457	571	650	9	3319	76
Lignite	24	14	42	0	0	80	2
Hydro	114	31	66	12	14	238	5
Nuclear	32	63	76	0	0	170	4
Gas, Naptha & Diesel	2	11	6	0	23	42	1
RES (Wind, Solar, Biomass & Others)	203	145	174	6	1	528	12
Total	1007	1721	935	667	47	4377	100

Share of RES in total generation (%)	20.11	8.43	18.62	0.83	2.24	12.07
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	34.58	13.86	33.84	2.61	32.72	21.38

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.018
Based on State Max Demands	1.033

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	200033	9:58	50
Non-Solar hr	182556	18:20	0

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES								
Import=(+ve) /Export =(-ve) for NET (MU)								
Date of Reporting: 09-Dec-2024								
Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	249	0.0	6.1	-6.1
3	765 kV	GAYA-VARANASI	2	57	1066	0.0	14.5	-14.5
4	765 kV	SASARAM-FATEHPUR	1	0	499	0.0	6.9	-6.9
5	765 kV	GAYA-BALIA	1	0	604	0.0	10.9	-10.9
6	400 kV	PUSAULI-VARANASI	1	0	170	0.0	3.5	-3.5
7	400 kV	PUSAULI -ALLAHABAD	1	0	124	0.0	2.4	-2.4
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	616	0.0	9.3	-9.3
9	400 kV	PATNA-BALIA	2	0	1044	0.0	18.6	-18.6
10	400 kV	NAUBATPUR-BALIA	2	0	718	0.0	5.9	-5.9
11	400 kV	BIHARSHARIFF-BALIA	2	22	283	0.0	5.2	-5.2
12	400 kV	MOTIHARI-GORAKHPUR	2	0	558	0.0	9.7	-9.7
13	400 kV	BIHARSHARIFF-VARANASI	2	0	0	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	94	0.0	0.1	-0.1
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.1	0.0	0.1
16	132 kV	GARWAH-RIHAND	1	30	0	0.5	0.0	0.5
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAUJI	1	0	0	0.0	0.0	0.0
ER-NR						0.6	93.1	-92.4
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	0	1529	0.0	22.6	-22.6
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	478	745	0.0	6.4	-6.4
3	765 kV	JHARSUGUDA-DURG	2	0	691	0.0	12.6	-12.6
4	400 kV	JHARSUGUDA-RAIGARH	4	0	713	0.0	11.4	-11.4
5	400 kV	RANCHI-SIPAT	2	45	274	2.9	0.0	2.9
6	220 kV	BUDHIPADAR-RAIGARH	1	14	94	0.0	0.9	-0.9
7	220 kV	BUDHIPADAR-KORBA	2	53	108	0.0	0.4	-0.4
ER-WR						2.9	54.4	-51.4
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	625	0.0	14.9	-14.9
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1990	0.0	38.3	-38.3
3	765 kV	ANGUL-SRIKAKULAM	2	0	2473	0.0	44.9	-44.9
4	400 kV	TALCHER-I/C	2	464	372	1.3	0.0	1.3
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	98.2	-98.2
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	233	0	3.6	0.0	3.6
2	400 kV	ALIPURDUAR-BONGAIGAON	2	527	0	7.2	0.0	7.2
3	220 kV	ALIPURDUAR-SALAKATI	2	96	0	1.5	0.0	1.5
ER-NER						12.3	0.0	12.3
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	668	0	16.2	0.0	16.2
NER-NR						16.2	0.0	16.2
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	3050	0.0	56.1	-56.1
2	HVDC	VINDHYACHAL B/B	-	0	307	0.0	7.2	-7.2
3	HVDC	MUNDRA-MOHINDERGARH	2	0	980	0.0	18.0	-18.0
4	765 kV	GWALIOR-AGRA	2	19	1690	0.0	19.6	-19.6
5	765 kV	GWALIOR-PHAGI	2	829	1685	1.7	19.5	-17.8
6	765 kV	JABALPUR-ORAI	2	0	700	0.0	21.3	-21.3
7	765 kV	GWALIOR-ORAI	1	869	0	13.3	0.0	13.3
8	765 kV	SATNA-ORAI	1	0	794	0.0	15.4	-15.4
9	765 kV	BANASKANTHA-CHITORGARH	2	1301	684	14.9	2.4	12.6
10	765 kV	VINDHYACHAL-VARANASI	2	0	1979	0.0	25.5	-25.5
11	400 kV	ZERDA-KANKROLI	1	274	90	2.9	0.3	2.6
12	400 kV	ZERDA -BHINMAL	1	189	62	2.0	0.2	1.8
13	400 kV	VINDHYACHAL -RIHAND	1	967	0	21.9	0.0	21.9
14	400 kV	RAPP-SHUJALPUR	2	666	148	4.3	0.4	3.9
15	400 kV	NEEMUCH-Chittorgarh	2	678	128	4.2	0.2	4.0
16	220 kV	BHANPURA-RANPUR	1	0	202	0.0	3.4	-3.4
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	1.7	-1.7
18	220 kV	MEHGAON-AURAIYA	1	113	0	1.3	0.0	1.3
19	220 kV	MALANPUR-AURAIYA	1	81	6	0.8	0.0	0.8
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						67.3	191.1	-123.8
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	996	0.0	14.2	-14.2
2	HVDC	RAIGARH-PUGALUR	2	0	3505	0.0	41.6	-41.6
3	765 kV	SOLAPUR-RAICHUR	2	1670	949	10.4	5.7	4.6
4	765 kV	WARDHA-NIZAMABAD	2	0	1830	0.0	28.4	-28.4
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2015	0.0	30.5	-30.5
6	400 kV	KOLHAPUR-KUDGI	2	1603	0	24.5	0.0	24.5
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	0	0.0	0.0	0.0
9	220 kV	XELDEM-AMBEWADI	1	0	124	2.3	0.0	2.3
WR-SR						37.2	120.5	-83.3
INTERNATIONAL EXCHANGES								
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)		
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	143	-60	20	0.47		
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	-111	0	-7	-0.18		
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-241	-150	-189	-4.55		
	NER	132kV GELEPHU-SALAKATI	-23	-13	-19	-0.45		
	NER	132kV MOTANGA-RANGIA	9	-4	4	0.10		
NEPAL	NR	NEPAL IMPORT (FROM UP)	-36	0	0	0.00		
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-36	0	-15	-0.36		
	ER	NEPAL IMPORT (FROM BIHAR)	0	0	0	0.00		
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-169	140	7	0.17		
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-494	-432	-451	-10.83		
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-766	-402	-638	-15.32		
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-56	0	-47	-1.12		

CROSS BORDER EXCHANGE SCHEDULE

Date of Reporting: 09-Dec-2024

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	5.04	0.00	0.00	0.59	0.00	0.00	5.63
Nepal	0.11	0.00	0.00	0.00	0.00	1.08	0.00	0.00	1.19
Bangladesh	11.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	11.21
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	11.32	0.00	5.04	0.00	0.00	1.67	0.00	0.00	18.03

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.72	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.72
Nepal	0.00	0.00	0.46	0.00	0.00	0.00	0.00	0.00	0.46
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	0.72	0.00	0.46	0.00	0.00	0.00	0.00	0.00	1.18

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.72	0.00	-5.04	0.00	0.00	-0.59	0.00	0.00	-4.91
Nepal	-0.11	0.00	0.46	0.00	0.00	-1.08	0.00	0.00	-0.73
Bangladesh	-11.21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-11.21
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-10.60	0.00	-4.58	0.00	0.00	-1.67	0.00	0.00	-16.85

Date of Reporting: 09-Dec-2024												
15 Min (INSTANTANEOUS) ALL INDIA GRID FREQUENCY, GENERATION & DEMAND MET (SCADA DATA)												
TIME	FREQUENCY (Hz)	DEMAND MET (MW)	NUCLEAR (MW)	WIND (MW)	SOLAR (MW)	HYDRO** (MW)	GAS (MW)	THERMAL (MW)	OTHERS* (MW)	NET DEMAND MET (MW)	TOTAL GENERATION (MW)	NET TRANSNATIONAL EXCHANGE (MW) (+ve) Import, (-ve) Export
		(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I=A-(C+D))	(J=B+C+D+E+F+G+H)	(K)
0:00	50.01	149486	6838	3828	19	8265	1746	129881	2759	145639	153336	-795
0:15	49.96	148815	6844	3956	20	7941	1773	129276	2724	144839	152534	-779
0:30	49.96	147604	6852	4017	19	7642	1783	128513	2741	143568	151567	-771
0:45	49.96	146982	6850	4013	16	7407	1782	128116	2782	142953	150966	-769
1:00	49.96	145922	6841	4055	7	7270	1776	127267	2764	141860	149980	-789
1:15	49.96	145207	6848	3990	2	6825	1787	126955	2820	141215	149227	-769
1:30	50.01	144038	6843	3942	-5	6812	1779	126001	2763	140101	148135	-779
1:45	49.96	143754	6841	3888	-1	6714	1793	125606	2809	139867	147650	-749
2:00	50.01	142505	6854	3838	-1	6671	1798	124746	2776	138668	146682	-766
2:15	50.01	141887	6854	3872	-2	6314	1793	124438	2830	138017	146099	-773
2:30	50.01	141459	6852	3867	-2	6276	1789	124146	2826	137594	145754	-777
2:45	49.96	141283	6851	3926	54	6207	1796	124000	2842	137303	145676	-787
3:00	50.01	141014	6858	3925	54	6151	1801	123831	2837	137035	145457	-798
3:15	49.96	141255	6850	3914	52	6555	1789	123635	2835	137289	145630	-755
3:30	50.01	140805	6858	4029	51	6886	1808	122968	2832	136725	145432	-809
3:45	50.06	141162	6864	4176	50	6981	1814	123066	2849	136936	145800	-826
4:00	50.06	141034	6854	4347	50	6941	1806	122919	2821	136637	145738	-800
4:15	50.06	142976	6863	4429	50	6896	1814	124670	2888	138497	147610	-817
4:30	50.06	144333	6875	4470	50	7131	1816	125958	2783	139813	149083	-865
4:45	50.06	145956	6877	4508	50	7406	1860	127167	2803	141398	150671	-847
5:00	50.06	147548	6857	4488	50	7557	1851	128805	2801	143010	152409	-932
5:15	50.01	150358	6858	4469	50	7628	1857	131358	2804	145839	155024	-930
5:30	50.06	153417	6859	4468	50	8212	1850	134122	2798	148899	158359	-898
5:45	50.06	156367	6858	4645	52	8900	1854	135894	2775	151670	160978	-918
6:00	50.16	158226	6865	4722	44	9044	1842	137334	2768	153460	162619	-739
6:15	50.01	161219	6868	4743	122	10236	1855	138661	2753	156354	165238	-738
6:30	50.09	165761	6848	4919	125	13102	1856	139998	2786	160717	169634	-783
6:45	49.99	169615	6858	5143	290	15085	1809	141375	2819	164182	173379	-787
7:00	49.99	172751	6858	5197	854	15928	1816	142891	2838	166700	176382	-774
7:15	49.89	176570	6858	5290	1876	16612	1817	144553	2857	169404	179863	-759
7:30	49.99	179477	6855	5245	3647	16779	1817	145647	2936	170585	182926	-744
7:45	49.99	181755	6867	5089	6480	16457	1811	145366	2938	170186	185008	-768
8:00	50.04	183532	6858	4930	10419	15879	1808	144019	2969	168183	186882	-759
8:15	50.04	186145	6864	4783	14958	13937	1800	144303	2930	166404	189575	-753
8:30	49.99	189101	6851	4632	19949	12507	1789	143795	2934	164520	192457	-692
8:45	49.99	191876	6856	4494	24809	11926	1754	142413	2956	162573	195208	-730
9:00	49.99	193540	6840	4319	29489	10386	1772	141285	2981	159732	197072	-791
9:15	49.79	196487	6851	3957	33464	9436	1761	141945	2999	159066	200413	-798
9:30	49.99	198778	6850	3745	37351	9272	1754	140658	3023	157682	202653	-803
9:45	49.99	199382	6859	3546	40363	9287	1751	138785	3046	155473	203637	-821
10:00	50.04	199779	6855	3506	43303	8719	1753	136675	3030	152970	203841	-863
10:15	49.99	198915	6862	3552	44785	8168	1746	134764	3036	150578	202913	-787
10:30	49.99	198041	6845	3413	46924	7270	1756	133069	3022	147704	202299	-936
10:45	50.11	197391	6860	3684	48272	6812	1727	131648	3036	145435	202039	-996
11:00	50.11	194606	6863	3827	48867	6725	1718	127981	3026	141912	199007	-981
11:15	50.11	193025	6829	3829	49866	6686	1717	125145	3073	139330	197145	-972
11:30	50.08	192689	6846	3583	51054	6570	1681	124597	3070	138052	197401	-1040
11:45	50.02	192265	6831	3443	51212	6622	1678	123612	3115	137610	196513	-1028
12:00	50.01	191158	6837	3328	51433	6742	1676	122469	3077	136397	195562	-996
12:15	50.06	188993	6819	3324	51764	6470	1671	120525	3071	133905	193644	-999
12:30	49.96	188137	6825	3098	51418	6302	1682	120122	3047	133621	192494	-1018
12:45	49.95	187285	6835	3327	50868	6506	1706	119078	3041	133090	191361	-997
13:00	49.95	184463	6843	3746	50307	6336	1678	116807	3018	130410	188735	-995
13:15	49.95	182159	6836	3601	49945	6065	1701	115338	3059	128613	186545	-1027
13:30	50.08	181743	6813	3361	48634	6605	1704	115971	3047	129748	186135	-1031
13:45	49.97	180802	6827	3359	47692	6822	1675	116202	2981	129751	185558	-975
14:00	50.02	180418	6811	3467	45907	6923	1661	117067	2953	131044	184789	-969
14:15	49.97	179938	6815	3635	44398	6697	1663	117877	2936	131905	184021	-980
14:30	49.92	180229	6813	3717	42064	6894	1680	120339	2916	134448	184423	-950
14:45	49.92	180208	6818	3676	39948	7058	1670	122212	2949	136584	184331	-1029
15:00	49.92	180121	6809	3760	35745	7187	1683	123617	2954	140616	181755	-1000
15:15	49.92	177805	6804	3887	33204	7538	1669	125158	2940	140714	181200	-997
15:30	50.02	177499	6806	3918	31083	7645	1700	127773	2905	142498	181830	-948
15:45	49.97	176499	6148	4190	27917	8267	1666	129407	2892	144392	180487	-906
16:00	50.07	175686	6139	4576	24645	8752	1666	130833	2887	146465	179498	-887
16:15	49.97	174051	6151	4879	20892	8838	1728	132614	2881	148280	177983	-896
16:30	49.97	174159	6137	4947	16515	8817	1669	137275	2861	152697	178221	-906
16:45	49.92	174977	6148	4973	12310	10484	1672	140605	2825	157694	179017	-886
17:00	50.02	175984	6137	4956	7572	11958	1679	144998	2761	163456	180061	-762
17:15	49.97	176048	6156	4990	4660	12590	1674	146987	2807	166398	179864	-741
17:30	50.07	177732	6151	5031	2497	14513	1673	148789	2883	170204	181537	-757
17:45	50.07	180293	6158	5212	1013	17754	1682	149777	2834	174068	184430	-737
18:00	50.07	180758	6156	5663	601	18507	1678	149321	2891	174494	184817	-744
18:15	50.02	182331	6160	6075	515	19565	1683	149500	2883	175741	186381	-725
18:30	50.02	181891	6163	6499	519	20057	1688	148142	2866	174873	185934	-714
18:45	50.02	180693	6147	6885	440	19491	1687	147078	2851	173368	184579	-706
19:00	50.02	178352	6146	7306	453	18840	1682	144726	2903	170593	182056	-736
19:15	50.07	175722	6338	7855	439	17992	1803	142157	2855	167428	179439	-720
19:30	50.02	172730	6164	8277	449	15424	1682	141583	2861	164004	176440	-731
19:45	50.02	170192	6156	8456	440	14215	1684	139961	2842	161296	173754	-700
20:00	50.07	167113	6171	8590	435	13045	1669	138018	2859	158088	170787	-732
20:15	50.02	164331	6162	8736	145	12297	1808	135827	2880	155450	167855	-704
20:30	49.97	162083	6163	8881	139	11750	1808	133924	2927	153063	165592	-718
20:45	49.97	160298	6164	9144	56	11225	1805	132733	2901	151098	164028	-686
21:00	50.12	157684	6163	9320	8	10919	1685	130619	2877	148356	161591	-660
21:15	50.02	155825	6174	9462	9	10103	1833	128748	2924	146354	159253	-684
21:30	49.97	153976	6169	9544	9	9718	1691	127373	2888	144423	157392	-792
21:45	50.02	151698	6183	9587	7	9217	1817	125870	2897	142104	155578	-810
22:00	50.07	149495	6162	9607	5	8980	1693	124110	2895	139883	153452	-838
22:15	49.97	149416	6173	9445	8	8861	1860	124140	2903	139963	153390	-796
22:30	49.97	148593	6174	9364	12	8946	1691	123462	2892	139217	152541	-786
22:45	49.97	147393	6189	9361	10	90						